Multimodal Deepfake Detection for Images, Video and Audio

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Group members

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Problem, scope, objectives

One local checker for images, video and speech, built as a documented path.

Image	**Video**	**Audio**
generated by GAN or diffusion	swapped or re-enacted faces	synthetic or converted speech

Objectives

1. Build one local system for images, video and speech.
2. Fix the evaluation protocol before any model.
3. Improve in steps, each from a written hypothesis; keep failed runs.
4. Report one finding per module with its limits, never one overall score.

Context, their protocol: Xception trained on FaceForensics++ c23 has a mean frame-level AUC of 0.9450 within the domain and 0.7718 across datasets (Yan et al., 2023).

Claimed, with intervals

- Face-video and speech detectors with sealed results
- An image probe, tested on older and newer generators
- A local platform with eight modules

Not claimed

- An overall verdict or fused score
- Audio-visual, pulse or generated-video detection
- Detection of modern generated images
- Any Gemini capability

Literature and the gap

Near 0.99 in-dataset, much lower across datasets; few papers report intervals or shortcut controls.

Literature (Appendix B, 40 papers)

- In-dataset FF++: video AUC about 0.99 (LipForensics).
- FF++ to Celeb-DF v2, video AUC: Xception 73.7, LipForensics 82.4, SBI 93.18, CLIP adapters (Effort, Forensics Adapter) 95.6 to 95.7.
- Newer methods (Celeb-DF++): every detector drops; 24-detector average 73.7.
- Audio-visual fusion work (FakeAVCeleb, LAV-DF) is mostly trained and tested on one dataset.
- Format and codec shortcuts inflate scores (Fake or JPEG, ECCV-W 2024).

Gap this project addresses

- Identity-disjoint splits with leakage and duplicate checks.
- Sealed tests read once, guarded and logged in a ledger.
- Three confirmatory tests declared before the read; Bonferroni.
- Group-bootstrap intervals, 3 seeds and a re-run Xception baseline.
- Shortcut controls on FakeAVCeleb, LAV-DF and modern images.
- A platform that reports every check with its tested scope.

Protocol and data

Splits are group-disjoint, sealed sets are read once, three tests were declared first.

Split once by source identity → **Train** fit the model → **Validation** temperature, threshold → **Development slice** exploratory only → **Sealed tests** read once, guarded

Task	Train and validate	Sealed tests
Face video	FaceForensics++ c23	FF++ test_in; Celeb-DF family (7,650 videos, 344 groups: 44 identities and 300 single YouTube videos); DFD
Speech	ASVspoof 2019 LA	ASVspoof test_in; In-the-Wild (42 speakers)
Image	ForenSynths ProGAN	ProGAN test_in; CNN-synthesis and diffusion sets

Why this protocol

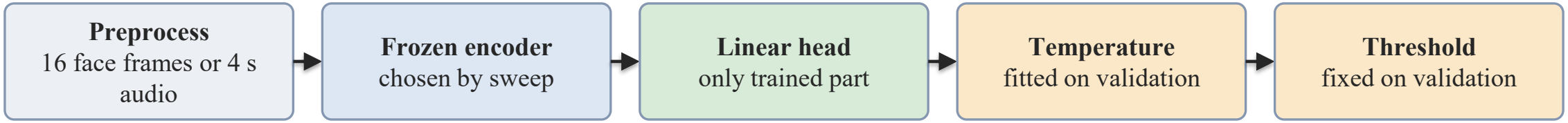
- No group on both sides of a split.
- A guard refuses a second read of a sealed set.
- Tuning uses only validation and the development slice.

Declared before any sealed read

- T1 video: final minus first version.
- T2 speech: final minus first version.
- T3 video: final minus re-run Xception.

Method: file to score

A file becomes a score: frozen encoder, linear head, temperature, threshold.



The mathematics, one line each

Clip score: s = sigmoid(logit(mean frame probability) / T)

Interval: 2,000 paired group-bootstrap resamples, 95%

T3 rule: non-inferior if the lower bound is above -0.0100

Encoder rule: ties with the leader, then the cheapest wins

What a score means

A score ranks files. The temperature is fitted on validation, so it is a probability only for content like validation. No output is the probability that a file is fake.

Working version, lessons

Face video underfits and speech collapses outside its corpus.

Task (frozen encoder)	Validation	Development slice
Face video (DINOv2 small)	0.7985	0.6933
Image (CLIP ViT-L/14)	1.0000	0.9935
Speech (HuBERT base)	0.9942	0.6100

ROC-AUC of the working version I-0. The development slice is exploratory, with wide intervals.

How the encoders were chosen

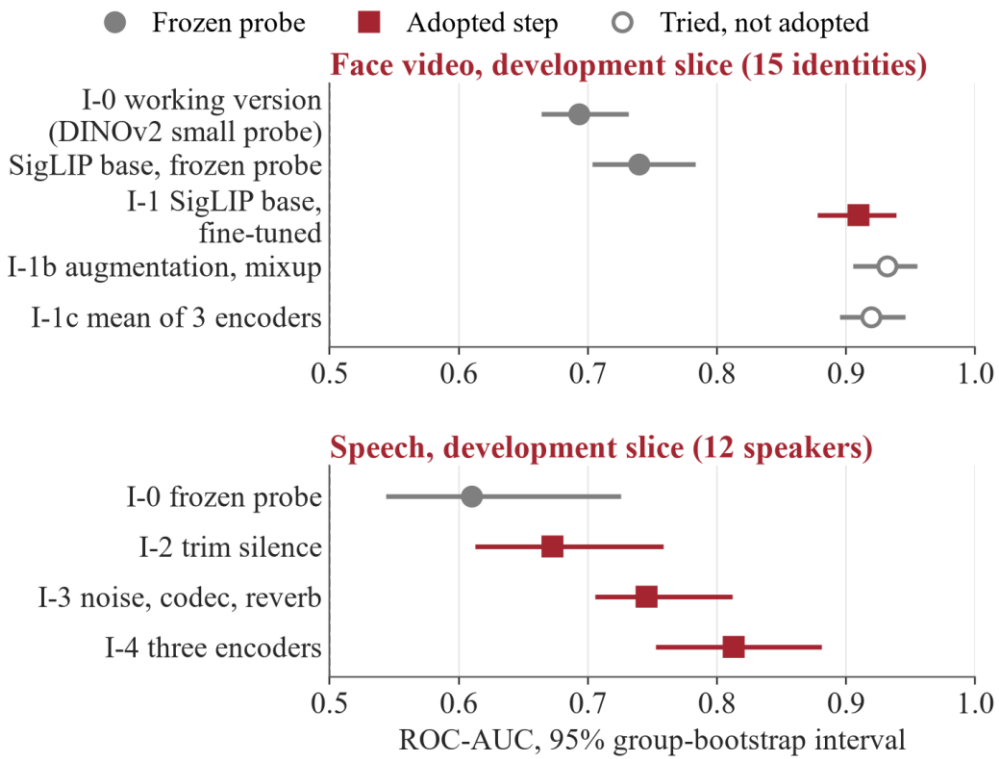
A measured sweep: the cheapest candidate that ties the leader wins.

What we learned

- Frozen face-video features underfit even the training data (training AUC 0.8367).
- Speech: 0.9942 on validation, 0.6100 on the development slice.
- Leading silence and level cues reverse sign between speech corpora.
- The temperature fitted on validation did not transfer.

Iteration ladder

A change entered only when it met its written success line.



Development slice (exploratory): face video, 15 identities; speech, 12 speakers. Hollow marks were tried and not adopted.

Reading

- Video: fine-tuning adds +0.1700 [+0.1228, +0.2114] over the frozen probe of the same encoder (exploratory).
- Speech: three steps, each met its line; point estimates rise, adjacent intervals overlap.
- Not adopted: mixup (a trade-off), three-encoder mean (no gain), LAV-DF fusion (a shortcut), pulse check (failed).

Sealed tests T1 to T3

T1 and T2 improved by the declared rule; T3 is non-inferior, not superior after Bonferroni.

Test	Sealed set	AUC, before to after	Paired difference [95%]	Reading
T1 video	Celeb-DF family, 7,650 videos, 344 groups	0.6238 to 0.8218	+0.1980 [+0.1688, +0.2242]	Improved by the declared rule; holds at 98.33%: [+0.1627, +0.2303]
T2 speech	In-the-Wild, 42 speakers	0.7516 to 0.9148	+0.1633 [+0.1177, +0.2266]	Improved by the declared rule; holds at 98.33%: [+0.1085, +0.2481]. Part of the gain removes a shortcut
T3 video vs re-run Xception	Celeb-DF family; Xception 0.8042 [0.7841, 0.8223]	0.8042 to 0.8218	+0.0176 [+0.0025, +0.0327]	Non-inferior at both levels (margin 0.0100). The 95% superiority rule was met but does not survive Bonferroni (98.33% interval [-0.0011, +0.0355]). The baseline ran at 224 px

- The 98.33% line was declared beside the 95% rule; a reader who wants family-wise control over the three tests reads the 98.33% column.
- T3 uses one baseline family at one input size; every sealed set was read once.

Speech: gain and shortcut

Part of the speech gain removes a shortcut; not all of it is better detection.

Sealed In-the-Wild AUC	I-0	I-2 trim, normalise	I-3 augment	I-4 three encoders
Point estimate (exploratory)	0.7516	0.8125	0.8672	0.9148

Single feature as fake score (AUC)	ASVspoof development	In-the-Wild development
Leading silence	0.375 [0.299, 0.453]	0.719 [0.425, 0.933]
RMS energy	0.594 [0.537, 0.649]	0.248 [0.073, 0.546]

- Trimming and normalising account for 0.061 of the 0.163 gain; a level cue remains: RMS AUC 0.135 [0.054, 0.382], development slice.
- Consistent with a sign-flipped shortcut; 12 speakers do not establish it.
- Test 2 shows better ranking on In-the-Wild, not how much is better detection of synthesis.

Image analysis

The ProGAN-trained probe misses 0.7395 of images from 21 recent generators.

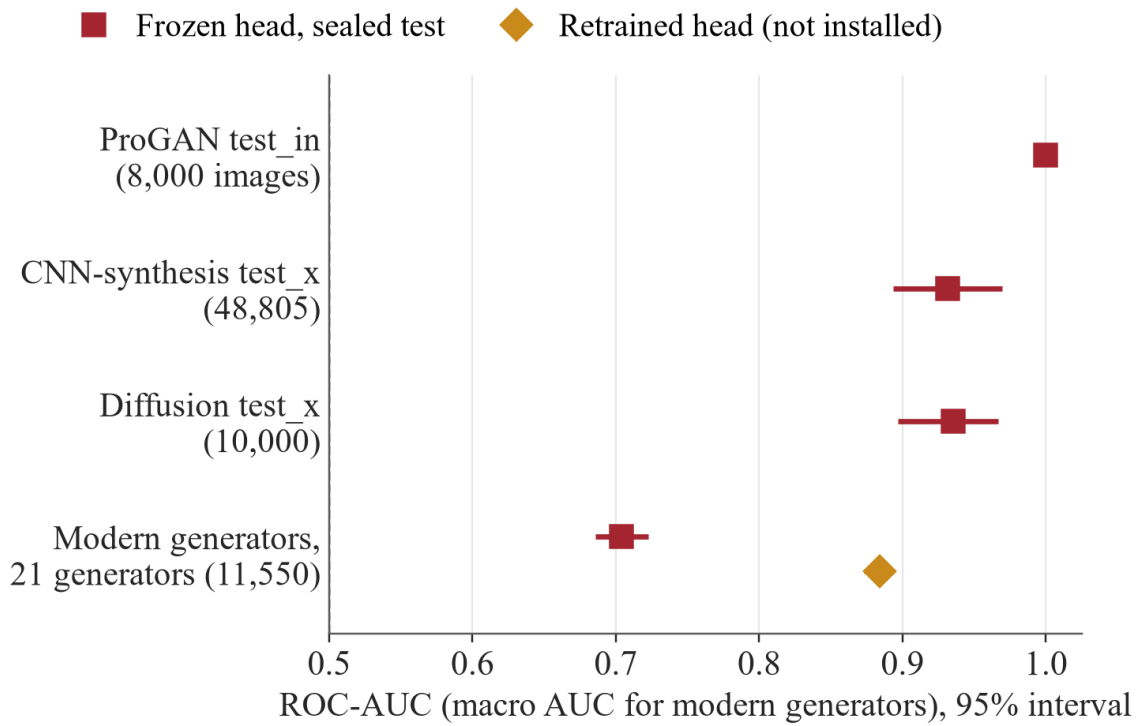
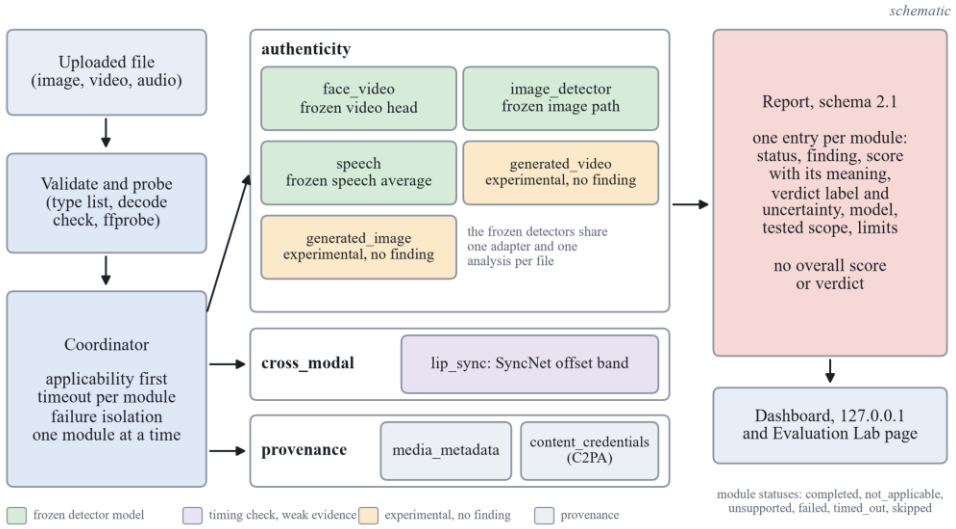


Image head, sealed test sets, each read once (exploratory). Last row: Community Forensics-Eval, macro AUC over 21 generators, same images for both heads.

- Frozen CLIP ViT-L/14 with a linear probe; faces use the video head.
- Older generators: AUC 0.9316 to 1.0000 (exploratory).
- Modern generators: macro AUC 0.7039; 0.7395 of generated images missed, 0.0418 of real flagged.
- A file-format cue alone reaches macro AUC 0.9301: the benchmark is confounded.
- Retrained head ranks better (0.8839) but flags 0.5213 of real images: not installed.

Platform and its limits

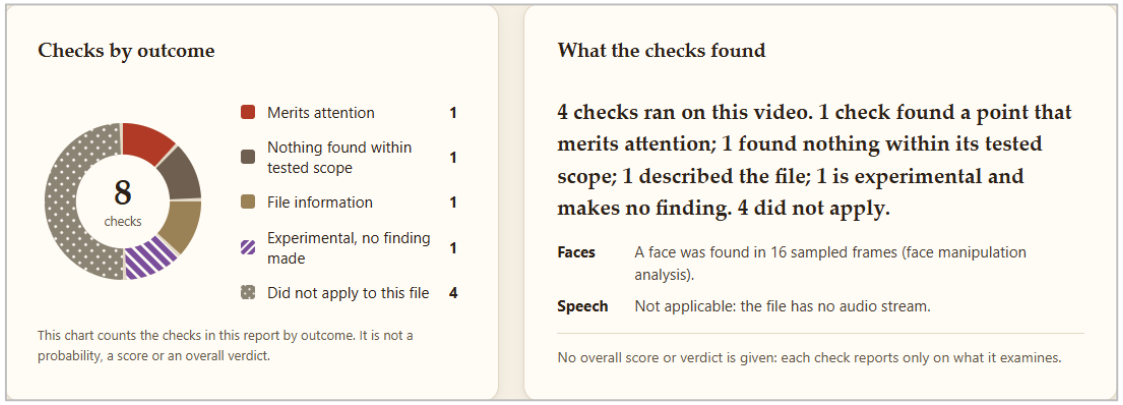
Eight modules and no overall score; the tests after the freeze limit what each may claim.

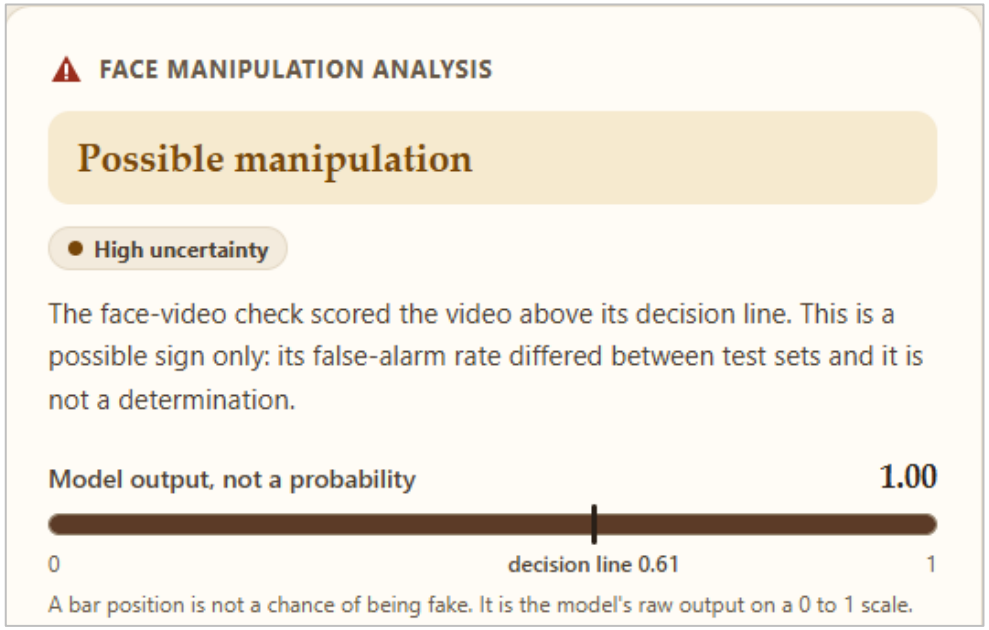


- Modules run one at a time; failures stay inside one module.
- Experimental heads give no finding.
- Software tests: 713 passed, 2 skipped (opt-in GPU tests), 13 deselected (data-dependent tests).

Check	What it allows
FakeAVCeleb	Criteria met (visual 0.9552, audio 0.9263), not evidence of detection quality: codec metadata alone gives 0.9985; about half of fake videos missed (sensitivity 0.480); thresholds do not transfer
LAV-DF	Visual part at chance within produced videos (0.5032, exploratory): no audio-visual claim
Generated video	At chance on DF26 (0.4957 [0.4755, 0.5168]): experimental, no finding
Modern images	Probe misses 0.7395; retrained head flags 0.5213 of real images: not installed

Dashboard





- Each check reports its own result, tested scope and limits; no overall score or verdict.
- The face-video flag reads Possible manipulation, with high uncertainty; no score is a probability.
- Experimental modules give no finding.

Gemini-generated images

No Gemini data, model or experiment exists; this is an open question, not a capability.

Status

- Only a feasibility assessment and a proposal exist.
- Nearest measured generator: Imagen 3, not Gemini.
- Gemini image models are not one model.

Imagen 3 in the benchmark (203 generated, 688 real)

Measure	Value
Frozen head, AUC	0.7691 [0.7352, 0.8020]
Frozen head, generated missed	0.8719
Retrained head, AUC	0.8646 [0.8360, 0.8917]
Retrained head, real flagged	0.5654
File-format cue alone, AUC	0.9731

Unresolved before any study

- Whether Google's terms allow training and testing a detector on Gemini outputs.
- Provenance of candidate sets: prompts, dates and model build are not recorded.
- Matched real images, and coverage of the different Gemini versions.

Comparison and limitations

Published numbers are context only; the limits are thresholds and dataset shortcuts.

Published numbers, context only

- Only the re-run Xception (T3) shares task, data and protocol.
- Speech EER, ASVspoof 2019 LA: ours 6.94% [5.94, 7.79]; AASIST 1.13% as quoted by its authors. AASIST was not re-run.
- In-the-Wild: ours 15.78% [12.75, 18.70]; full-set published 33.9 and 7.2%. Different test set, no ranking.
- DF26: ours at chance, 0.4957 [0.4755, 0.5168]; authors report 44.0 to 69.7 without intervals. Training data differ, no ordering.

Limitations

- Thresholds do not transfer. Post hoc false-alarm rate (re-analysis of sealed scores, not confirmatory): 0.1098 on Celeb-DF family reals, 0.1405 on DFD.
- Shortcuts inflate AUC: codec, production cue, file format.
- Small development slices; T3 used one baseline family at 224 px.
- Not tested: edited images, social-media recompression.

Conclusion and demo

Conclusion

- Group-disjoint protocol; sealed sets read once; three tests declared first.
- Video: Celeb-DF family AUC 0.8218 against 0.6238 (+0.1980 [+0.1688, +0.2242]). Non-inferior to a re-run Xception (margin 0.0100); the 95% superiority rule was met but does not survive Bonferroni; baseline at 224 px.
- Speech: In-the-Wild 0.9148 against 0.7516 (+0.1633 [+0.1177, +0.2266]); part of the gain removes a shortcut.
- Limits: the image probe misses 0.7395 of recent generators; no audio-visual claim (visual part at chance within produced videos, 0.5032); FakeAVCeleb criteria met but not evidence of detection quality (codec 0.9985); thresholds do not transfer.

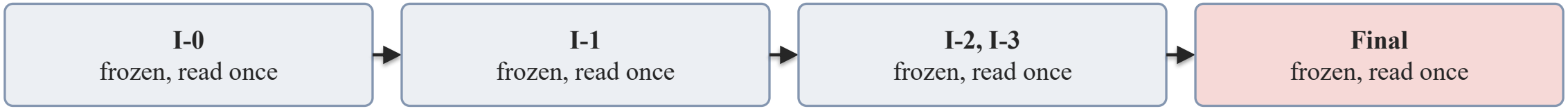
Demo summary

- Local dashboard on 127.0.0.1; uploads deleted.
- Live check: 8 of 8 demonstration inputs as specified.
- Tests: 713 passed, 2 skipped (opt-in GPU tests), 13 deselected (data-dependent tests).
- Read each result as a possible sign within its tested scope.

Backup slides

- B1 Statistics and decision rules
- B2 Preprocessing and parity
- B3 Encoder shortlist and sweep
- B4 I-0 diagnosis
- B5 I-1b and I-1c
- B6 LAV-DF control and pulse check
- B7 Generated video, lip sync, modern generators
- B8 Platform detail
- B9 Licences, data balance, environment
- B10 Sealed AUC by test set

B1 Statistics and rules



Metrics

- Headline: ROC-AUC on a dataset never used in training, 95% group-bootstrap interval.
- Also EER, balanced accuracy at the validation threshold, ECE with 15 bins.
- 2,000 resamples, percentile interval, groups resampled with replacement, paired.

Decision rules

- Non-inferior: lower end of the paired interval above -0.0100 (set beforehand; the floor of 0.01 decides).
- Superiority needs the whole interval above zero; for T3 it holds at 95% but not after Bonferroni.
- Encoder: the tie set against the leader, then the cheapest wins.
- Ledger of 74 entries, no repeat read.

B2 Preprocessing, parity

Input	Rule
Video	16 uniformly spaced frames; YuNet face detector; largest face enlarged by 1.3; 224 by 224 crop, JPEG quality 95; frames without a face are left out and reported
Image	A face crop if a face is found, otherwise the whole image processed by the encoder's own rule
Audio	First 64,600 samples at 16 kHz, repeat-padded; silence trimmed and peak normalised (final system)
Consistency	One shared package for training, evaluation and inference; a configuration hash is stored in every checkpoint and checked at load

Parity

Scores from stored features and from raw files agree within 0.0019 (tolerance 0.02). The check shows that the path works, not accuracy.

B3 Encoders and sweep

Task	Tie set with the leader	Selected, development AUC	Leader
Face video, 15 identities	DINOv2 small and base, SigLIP base, both CLIP ViT-L/14	DINOv2 small 0.6933 [0.6639, 0.7314]	CLIP ViT-L/14 OpenAI 0.7464 [0.6721, 0.8306]
Image, 2 generator units	CLIP ViT-L/14 OpenAI only	CLIP ViT-L/14 OpenAI 0.9935 [0.9803, 0.9982]	same
Speech, 12 speakers	HuBERT base, WavLM base plus, Whisper small	HuBERT base 0.6100 [0.5434, 0.7256]; inconclusive	Whisper small 0.6380 [0.5338, 0.7700]

Encoder	Parameters	Seconds per 1,000 inputs	Weights licence
DINOv2 small	22.1 M	0.94	Apache-2.0
SigLIP base	92.9 M	1.44	Apache-2.0
CLIP ViT-L/14 (OpenAI)	304.0 M	4.45	research use (accepted)
HuBERT base	94.4 M	2.04	Apache-2.0
WavLM base plus	94.4 M	2.53	research use (accepted)
Whisper small encoder	88.2 M	11.53	Apache-2.0 / MIT

B4 I-0 diagnosis

Task	Observations (battery)	Candidate diagnoses
Face frames	Train 0.837, validation 0.799, development 0.693 (gap +0.105). Weakest on validation: NeuralTextures 0.697, FaceSwap 0.778. Development ECE 0.205 before and 0.222 after temperature scaling.	Bias first: the frozen feature separates even the training data only moderately; then shift.
Images	Train and validation 1.000, development 0.994 on 2 generator units. ECE 0.046 before and 0.005 after.	No bias or variance visible; shift to other generators cannot be measured with 2 units.
Speech	Validation 0.994, development 0.610 (gap +0.384). Real In-the-Wild speech scored as fake. ECE 0.329 before and 0.405 after.	Shift is dominant; a missing signal or a shortcut is a candidate; calibration worsens under shift.

Exploratory, development slice. Diagnoses only: no lever is chosen here. No sealed set was read for them.

B5 I-1b and I-1c

Face video, 3 seeds	Validation AUC	Development AUC	Validation, paired with I-1	Slice, paired with I-1
I-1, fine-tuned SigLIP base	0.9899 [0.9809, 0.9967]	0.9096 [0.8776, 0.9388]	reference	reference
I-1b, augmentation and mixup	0.9705 [0.9618, 0.9795]	0.9315 [0.9055, 0.9548]	-0.0194 [-0.0251, -0.0129]	+0.0219 [+0.0090, +0.0382]
I-1c, mean of three encoders	0.9903 [0.9823, 0.9968]	0.9192 [0.8948, 0.9457]	+0.0004 [-0.0011, +0.0023]	+0.0096 [-0.0020, +0.0231]

I-1b: does not enter

Slice up, validation down by more than the allowed 0.0079: a trade-off.

I-1c: does not enter

The slice interval includes zero; no gain it can show.

B6 LAV-DF and pulse

LAV-DF validation, exploratory	Own label, all videos	Produced videos only	Originals vs audio-only fakes
Visual head, trained on LAV-DF	0.7491 [0.7280, 0.7697]	0.5070 [0.4771, 0.5361]	0.9912 [0.9864, 0.9951]
Audio head, trained on LAV-DF	0.8604 [0.8435, 0.8769]	0.8574 [0.8394, 0.8752]	0.8711 [0.8474, 0.8920]
Frozen face head, FF++	0.3838 [0.3599, 0.4079]	not applicable	0.2876 [0.2568, 0.3174]

LAV-DF sealed subset, exploratory	Fused label, 2,000 videos	Produced videos, 1,500
Visual head, trained on LAV-DF	0.9921 [0.9880, 0.9956]	0.5032 [0.4746, 0.5320]
Audio head, trained on LAV-DF	n/a	0.8480 [0.8281, 0.8670]

- No audio-visual detection is claimed: the visual part is at chance within produced videos.
- Pulse check fails: 0.301 [0.260, 0.346] above the shuffled control, 0.6 required; positive control recovers 0.008 amplitude in 0.873 of clips.

B7 Video, lip sync, modern

Generated-video head	DeepAction development	DF26 held out
Temporal geometry	0.5073 [0.4608, 0.5542]	0.5077 [0.4723, 0.5424]
Mean embedding, logistic regression (installed)	0.8281 [0.7908, 0.8662]	0.4957 [0.4755, 0.5168]
Seven pixel statistics (control)	0.7613 [0.7182, 0.8033]	0.5604 [0.5361, 0.5852]

Head trained on DF26 open generators	Result [95% interval]	Reading
AUC, open families held out	0.7453 [0.7174, 0.7710]	chosen head
AUC, commercial generators	0.7548 [0.7189, 0.7885]	one counted look
Real FF++ and LAV-DF clips flagged	0.8023 [0.7625, 0.8368]	confound persists

Lip-sync rule	Real clips flagged	Wav2Lip clips flagged
Tolerance of 2 frames, fresh clips	15.0 and 13.3 percent	85 to 93 percent
Band of -6 to +2 frames, disjoint clips	1.3 percent [0.4, 3.8]	0 of 160

A head trained on DF26's open generators reached 0.7548 on the commercial generators (one counted look); the installed head was at chance there; the two used different real sets.

B8 Platform detail

Report contract and verdicts

- Sections: authenticity, cross-modal, provenance; each module appears exactly once.
- Each module: label, tier, raw score with its kind, uncertainty, tested scope; is_probability is always false.
- The contract checker rejects any overall, fused or verdict field.
- The face-video flag carries high uncertainty because its false-alarm rate differed between test sets.

Evaluation Lab and tests

- Re-analyses recorded score files; runs no model, writes no ledger entry.
- A sealed split only as a re-analysis of a ledger entry with a matching SHA-256.
- Tests: 713 passed, 2 skipped (opt-in GPU tests), 13 deselected (data-dependent tests).
- Live browser check: 8 of 8 demonstration inputs behaved as specified.

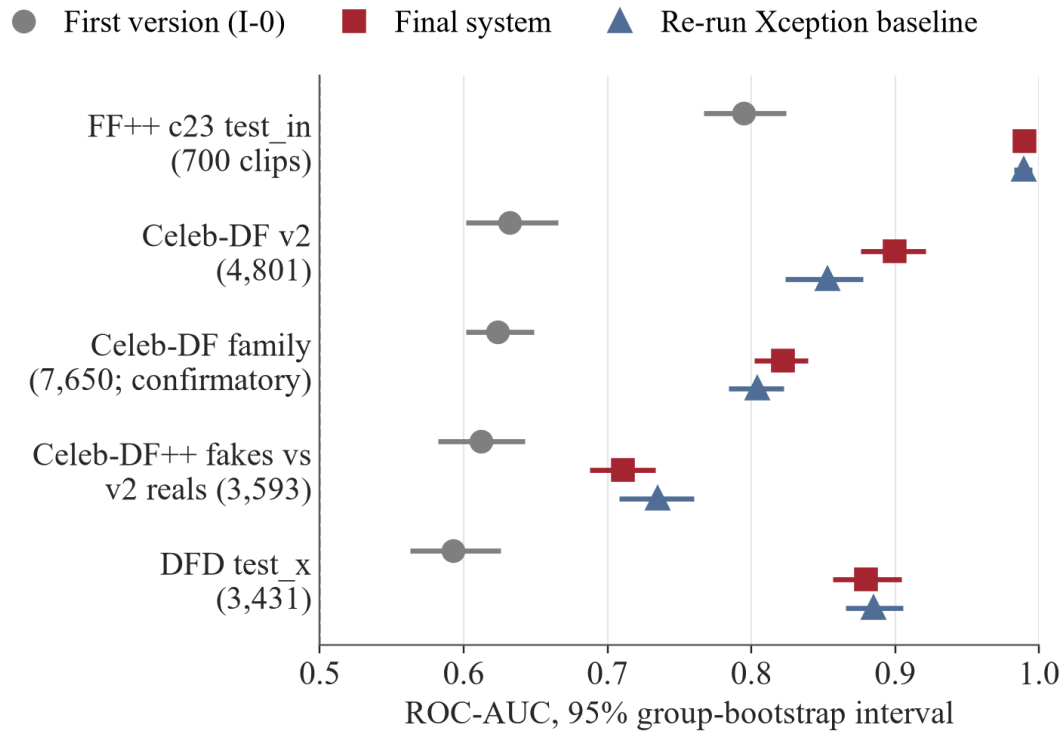
B9 Licences, balance, setup

Data or weights	Terms as recorded	Consequence
FaceForensics++	Non-commercial research and education only	Training; no frames shown
Celeb-DF v2, Celeb-DF++	Non-commercial research; no publication of derived data	Sealed tests; no per-video score shown
ASVspoof 2019 LA	ODC-By (README)	Audio training and validation
OpenAI CLIP, WavLM weights	No clean licence statement; conditional	Research-only; stated as a limitation

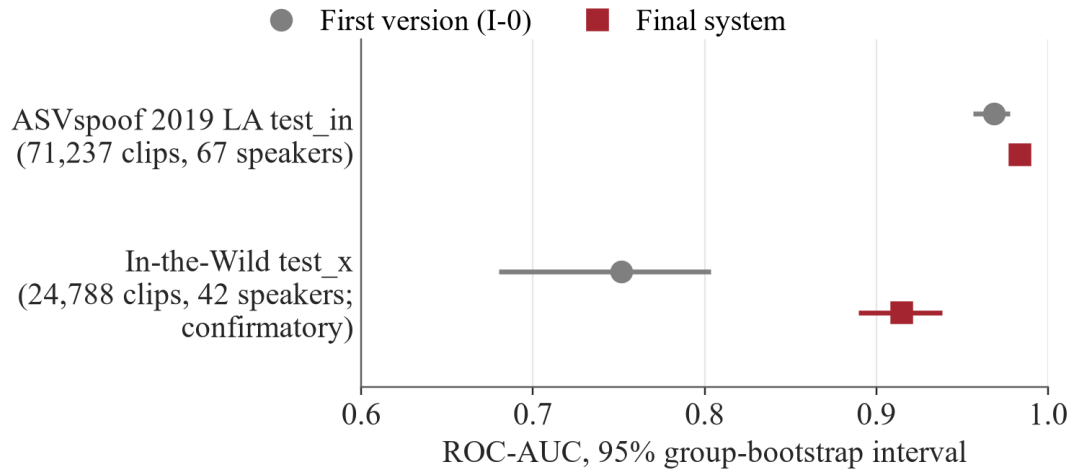
Family test	Real	Fake	Groups
Celeb-DF v2	744	4,057	344
Celeb-DF++ new	0	2,849	44 identities
Pooled	744	6,906	344

Item	Measured value
GPU	RTX 5070, 12 GB, driver 595.79
CPU, RAM	Ryzen 9 7900X, 31 GB
Software	Windows 10 Pro; Python 3.13.15; PyTorch 2.14.1, CUDA 13.0

B10 Sealed AUC by set



Face video, sealed, read once. Only the Celeb-DF family is confirmatory.



Speech, sealed, read once. In-the-Wild is confirmatory; ASVspoof is exploratory.